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RESEARCH ARTICLE

MammoClean: Toward Reproducible and Bias-Aware AI in Mammography Through Dataset Harmonization

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ABSTRACT The development of clinically reliable artificial intelligence (AI) systems for mammography is hindered by profound heterogeneity in data quality, metadata standards, and population distributions across public datasets. This heterogeneity introduces dataset-specific biases that can compromise the generalizability of AI models, a fundamental barrier to clinical deployment. We present *MammoClean*, a public framework for dataset standardization and bias identification in mammography datasets. *MammoClean* standardizes case selection, image processing (including laterality and intensity correction), and unifies metadata into a consistent multi-view structure. We provide a targeted review of breast anatomy, imaging characteristics, and public mammography datasets to systematically identify key sources of heterogeneity. We apply *MammoClean* to three large-scale heterogeneous datasets with detailed annotations (CBIS-DDSM, TOMPEI-CMMD, VinDr-Mammo) as a focused case study to quantify distributional shifts and illustrate the impact of data corruption on AI model performance. Applying *MammoClean* to the three datasets, we quantify substantial distributional shifts in breast density and abnormality prevalence. We further illustrate the direct impact of data corruption: AI models trained on corrupted datasets exhibit notable performance degradation compared to their curated counterparts. By using *MammoClean* to identify and document sources of heterogeneity, researchers can more reliably prepare unified multi-dataset training corpora as a foundation for developing robust models. *MammoClean* provides a reproducible pipeline for bias-aware dataset preparation in mammography, facilitating fairer comparisons across studies. The open-source code will be publicly available from: <https://github.com/Minds-R-Lab/MammoClean>

INDEX TERMS Mammography, breast cancer, data harmonization, dataset bias, deep learning.

I. INTRODUCTION

Breast cancer is the most common cancer among women and a leading cause of cancer deaths [1]. Regular screening with mammography is the most effective tool to detect breast cancer in early stages and prevent breast cancer-related

deaths [2]. Therefore, mammography is the primary imaging tool for breast cancer screening and diagnosis, and it is also commonly used in follow-up [3], [4] (see Figure 1). The time-consuming interpretation time of mammograms, combined with the low prevalence of cancer in mammography images, makes deep learning models, an important subset of Artificial Intelligence (AI), a promising approach for automating parts of the clinical workflow and reducing the burden on

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healthcare professionals. However, a critical challenge for developing AI systems for mammography is the profound heterogeneity across publicly available datasets in terms of data quality, metadata standards, annotation styles, and population distributions. These differences introduce dataset-specific biases that can undermine the reproducibility and generalizability of AI models, motivating the need for systematic dataset harmonization.

Mammography uses low-dose X-rays to visualize breast tissue and underlying abnormalities, and the features of the resulting images vary depending on the imaging technique and the specific view captured. Among the available technologies, Digital Mammography (DM) or Full-Field Digital Mammography (FFDM) are the most widely used for producing two-dimensional (2D) mammographic images [5]. FFDM captures the entire breast in high-resolution grayscale images, providing faster interpretation compared to traditional Screen-Film Mammography (SFM) [6]. As a result, FFDM has largely replaced SFM in clinical practice.

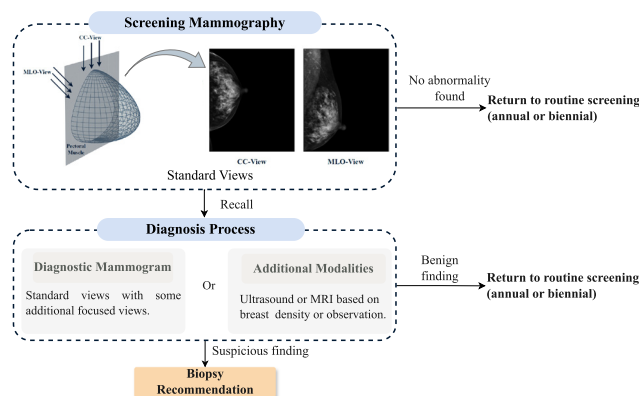


FIGURE 1. The role of mammography imaging in breast cancer screening and diagnosis.

The main limitation of FFDM lies in projecting the three-dimensional (3D) breast into 2D images, which inevitably causes information loss and tissue overlap. This is one of the main limitations of mammography and can cause early-stage cancers to be obscured behind dense and heterogeneous normal breast tissue. This challenge can be partially mitigated by acquiring and interpreting multiple views of the breast, which improves visualization of the 3D structure. Digital Breast Tomosynthesis (DBT), also known as 3D mammography, directly addresses this issue by capturing multiple angled images and reconstructing them into high-resolution 3D views [7]. DBT significantly improves cancer detection and reduces false positives; however, interpretation of DBT images generally requires more time compared to FFDM. Contrast-enhanced mammography (CEM) is another imaging method that uses a contrast agent to increase the visibility of abnormalities [8]. By having a low-energy image similar to the standard mammogram and a high-energy image with contrast, the radiologist can analyze a more detailed view and can better detect and characterize abnormalities.

One of the most comprehensive approaches to mammography image analysis is the multi-view strategy. For each breast, two standard projections are acquired: the Cranio-Caudal (CC) view, captured from above the breast, and the Medio-Lateral Oblique (MLO) view, obtained from the side at an angle. Radiologists typically employ two main strategies, contralateral comparison and ipsilateral correlation, to identify corresponding findings across both views and to assess differences between the same views of the left and right breasts. This approach enables more reliable confirmation of suspicious findings and represents the most thorough method for interpreting a complete mammographic study.

Several studies have demonstrated encouraging outcomes from applying AI, and particularly deep learning, to mammography analysis. Reported results include cancer detection rates comparable to double reading without an increase in recalls [9], an improvement of up to 4% in detection when replacing one radiologist with AI in double-reading setups [10], and reduced recall rates for low-risk cases [11]. Several studies have developed and evaluated deep learning models for mammography analysis [12], [13], [14], [15], [16], [17], leveraging publicly available or private datasets.

A critical requirement for the development of AI-based models in mammography image analysis is the availability of standardized and harmonized data, which enhances both the training process and overall model performance. However, most datasets are collected, annotated, and formatted under heterogeneous protocols, resulting in inconsistencies in image quality, labeling standards, and metadata representation. Such variability creates challenges for reproducibility, interoperability, and ultimately the generalizability of AI models across diverse clinical environments. Well-processed and harmonized datasets not only enable fair comparisons between methods but also provide a reliable foundation for developing robust models capable of handling real-world diversity, while unnecessary variability is reduced through harmonization. Therefore, addressing dataset heterogeneity and ensuring access to standardized, curated, and bias-aware resources is essential for advancing the clinical applicability of AI models.

Some studies have explored harmonization and pre-processing techniques for mammography images; however, these efforts have primarily focused on contrast enhancement methods aimed at improving the visibility of abnormalities or anatomical structures [18], [19], [20], [21], [22]. Notably, most of these techniques require parameter adjustments, which may vary across clinical environments and patient populations, thereby limiting their generalizability. Other critical technical aspects of harmonization are often overlooked or only vaguely addressed in existing models. Reference [23] proposed a framework for harmonizing breast cancer datasets from five public repositories, but their work was primarily focused on Magnetic Resonance Imaging (MRI), leaving mammography-specific challenges insufficiently addressed. Recently, vision-language models

have been proposed to leverage heterogeneous clinical data by pairing mammography images with radiology reports to improve breast cancer detection, even generating synthetic reports when textual data are unavailable from the provided abnormality annotations [24]. However, our work takes a complementary direction by directly addressing the underlying issue of dataset heterogeneity and bias. A key gap in the existing literature is the absence of a comprehensive, publicly available framework that integrates case selection, quality control (including laterality and intensity correction), metadata harmonization, and systematic bias quantification for mammography datasets. While individual preprocessing steps are well known, no existing tool provides their unified, reproducible, and extensible integration specifically tailored for mammography.

The primary identity of this paper is a harmonization framework contribution, supported by a targeted review of mammography fundamentals and datasets that motivates and contextualizes the framework design. We begin by introducing key terminologies to establish a clear understanding of the datasets, their inherent abnormalities, and the nature of these variations. Building on this, we review available datasets, focusing on those that are publicly accessible or obtainable upon request, and analyze their characteristics in detail. A critical challenge facing current mammography AI models is their limited generalization to unseen datasets. We hypothesize that quantifiable dataset biases are a contributing factor to this phenomenon. To address this, we present *MammoClean*, a harmonization framework that systematically integrates known harmonization needs into a unified, extensible, and publicly available pipeline, going beyond individual preprocessing steps by incorporating case selection, quality control, metadata unification, and bias quantification. Applied to three public datasets as a focused case study, the pipeline includes detailed technical considerations for robust dataset preparation and is released publicly to enhance community adoption and reproducibility. Finally, we conduct evaluations to identify and quantify sources of heterogeneity and bias that could undermine model performance across different datasets or in real-world clinical settings.

This paper may be particularly useful for researchers and developers in medical imaging, as well as for clinicians and dataset curators. For researchers and AI model developers, it offers perspectives on the intrinsic properties of mammography images and their associated abnormalities, which could support the design of models that are more robust to population-specific differences and less sensitive to case-selection biases. Clinicians might gain insights into how imaging variations and dataset composition can influence the diagnostic performance of AI models, even in ways that may not directly affect radiologists. Dataset curators and organizers may also find suggestions on what types of additional information could be considered in future datasets to enhance their research and clinical relevance.

By attempting to bridge technical and clinical perspectives, this work aims to contribute to the ongoing efforts toward more reliable, fair, and clinically meaningful AI-driven solutions for breast cancer detection and diagnosis.

In summary, the core problem addressed in this paper is the lack of a unified, reproducible framework for harmonizing heterogeneous mammography datasets and systematically identifying the biases they introduce. Our proposed solution, *MammoClean*, fills this gap by providing an open-source, modular pipeline that standardizes imaging data and metadata, performs quality control, and quantifies distributional biases across datasets. The main contribution is not a fundamentally new algorithm, but rather the practical, systematic integration and operationalization of known harmonization needs into a publicly available tool that enables the research community to build more consistent and bias-aware AI systems for mammography. Our contributions are summarized as follows:

- We provide a comprehensive review of mammography imaging fundamentals. This includes breast anatomy, tissue texture characteristics, and detailed analysis of main findings and their radiological presentations. We follow this with an in-depth survey of publicly available mammography datasets. This survey reveals substantial heterogeneity in their characteristics, metadata structures, and annotation styles.
- We propose *MammoClean*, a public pipeline that integrates and operationalizes known harmonization needs into a unified framework for dataset harmonization. The pipeline offers three key capabilities. First, it provides extendibility through a modular, open-source architecture adaptable to new datasets. Second, it enables bias identification that systematically documents and measures dataset heterogeneity across clinical decisions, breast density distributions, and abnormality prevalence. Third, it ensures reproducibility through fully documented and standardized processing steps, enabling consistent dataset preparation across studies.
- We conduct a focused case study evaluating three harmonized public datasets, revealing key differences in clinical characteristics. This evaluation illustrates how standardization can facilitate fairer model comparisons and more robust development strategies for AI-driven mammographic image analysis.

II. MAMMOGRAPHY ANALYSIS

In this section, we first describe breast anatomy and tissue composition as a key factor influencing both the visual appearance of mammography images and the associated risk and difficulty of abnormality detection. We then discuss the main categories of abnormalities typically observed in mammograms, followed by a brief overview of the clinical workflow.

A. BREAST ANATOMY AND TISSUE COMPOSITION

The anatomy of the breast is primarily composed of fibroglandular tissue (lobes and lobules), adipose tissue, milk ducts, and blood and lymphatic vessels, situated on the pectoral muscle. Figure 2 illustrates the anatomy of the breast and its projections in different mammographic views. The proportion of fibroglandular to adipose tissue varies among individuals, leading to different breast densities, which directly affect both the appearance of mammography images and the difficulty of detecting abnormalities. The Breast Imaging-Reporting and Data System (BI-RADS) provides a standardized four-category scale for classifying breast density [25] (Table 1); Figure 3 illustrates visual differences across these categories. Importantly, both tumors and dense breast tissue exhibit similar radiographic intensities, making their differentiation particularly challenging for radiologists and AI-based systems alike.

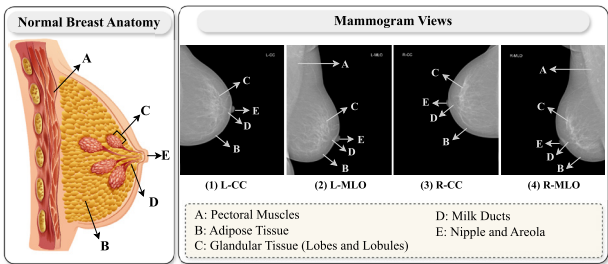


FIGURE 2. Illustration of breast anatomy and its appearance across different mammographic views, images are from [26].

TABLE 1. BI-RADS scaling for breast density, based on 5th edition [25].

BI-RADS	Type
A	Almost entirely fatty
B	Scattered areas of fibroglandular density
C	Heterogeneously dense
D	Extremely dense

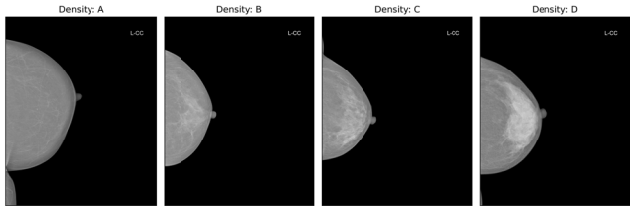


FIGURE 3. Different breast density categories on left CC mammography images from normal cases, images are from [26] (A: almost entirely fatty, B: scattered areas, C: heterogeneously dense, and D: extremely dense).

Breast density is influenced by several biological and demographic factors. While breast density varies between genders, the focus of screening mammography is primarily on female patients. It is inversely correlated with age, with younger individuals typically presenting higher density due to hormonal differences [27]. Population-based studies also

show significant ethnic variation; for instance, dense breast tissue is more prevalent among Asian women compared to White women [28]. These density-related distributional differences across populations are a key source of dataset bias that harmonization frameworks such as *MammoClean* aim to identify and quantify, as they directly affect both model training and cross-dataset generalization. Figure 4 extends the schematic of [29] to illustrate how patient-level factors such as age, sex, and ethnicity influence both disease-related and non-disease-related imaging features.

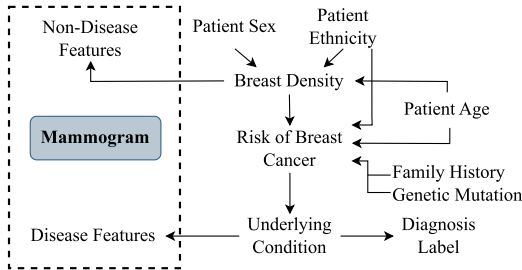


FIGURE 4. Causal relationships between various factors and their impact on mammography images for both disease-related and non-disease-related features. Patient sex, ethnicity, and age are common factors influencing breast density, which directly affects non-disease visual appearances in images and, by altering breast cancer risk and detection difficulty, also impacts disease-related features.

B. COMMON MAMMOGRAPHIC FINDINGS

Mammographic findings are generally classified into four main categories: masses, calcifications, architectural distortion, and asymmetry, with additional secondary signs such as skin thickening, skin retraction, and nipple retraction also observable. The primary categories are summarized as follows:

Masses are space-occupying lesions analyzed based on shape, margin, and density [30], [31]. Benign masses (e.g., cysts, fibroadenomas) tend to be oval or round with circumscribed margins and low density, while malignant masses are frequently irregular, spiculated, and of higher density. See Figure 5 for a visualization of different mass types.

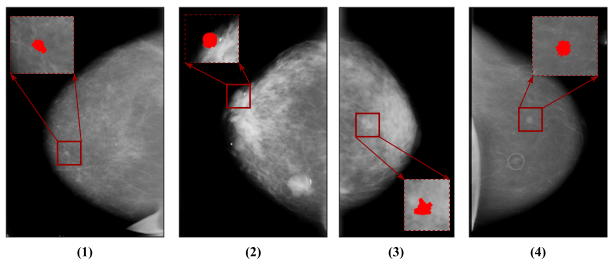


FIGURE 5. Different mass types, images are from [32]: (1) round with circumscribed margins (BI-RADS 2, benign), (2) oval with circumscribed margins (BI-RADS 3, benign), (3) oval with ill-defined margins (BI-RADS 4, malignant), and (4) irregular shape with spiculated margins (BI-RADS 5, malignant).

Calcifications are tiny calcified deposits appearing as bright dots in mammograms, characterized by morphology

and distribution [30]. Typically benign types include skin, vascular, and coarse calcifications, while suspicious types are small (<0.5 mm), with clustered, linear, or segmental distributions. Detecting such fine-grained patterns in high-resolution images remains a key challenge for AI models, as most approaches resize images to lower resolutions, risking loss of subtle detail. See Figure 6 for examples.

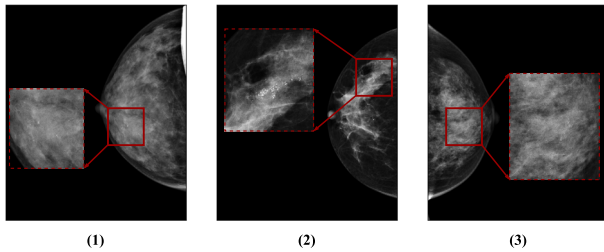


FIGURE 6. Different calcification types, images are from [33], [34]: (1) small round calcifications with scattered distribution throughout the whole breast (BI-RADS 2, benign), (2) pleomorphic calcifications with segmental distribution in the medial breast (BI-RADS 5, malignant), and (3) amorphous, indistinct calcifications with grouped distribution in the medial breast (BI-RADS 3, malignant).

Architectural Distortion is defined as a disruption of the normal breast architecture without the presence of a discrete mass [35]. It may occur as a result of post-surgical or post-therapeutic changes. However, when no benign cause is identified, it is considered highly suspicious for malignancy and requires careful evaluation.

Asymmetry describes an area of increased fibroglandular density differing from the contralateral breast. BI-RADS defines four subtypes [25]: asymmetry (single projection only), focal asymmetry (visible on at least two projections), global asymmetry (involving at least one quadrant), and developing asymmetry (new or increasing over time), with the latter considered more clinically significant.

C. CLINICAL WORKFLOW

In clinical practice, mammography may be performed for screening, diagnostic evaluation, or follow-up. The resulting report includes breast density and a description of all observed findings, culminating in a BI-RADS assessment (Table 2). Clinical management then follows one of four pathways: additional imaging, routine screening, short-term follow-up, or biopsy, with histopathological examination serving as the gold standard for malignancy confirmation.

An important distinction exists between the BI-RADS assessment and the final histopathological diagnosis. The BI-RADS score reflects the radiologist's impression based on imaging characteristics and represents a probabilistic assessment of malignancy risk, whereas the biopsy-confirmed diagnosis provides a definitive determination of the pathological nature of a finding. This distinction is critical for AI model development: models trained to predict BI-RADS scores are learning to replicate a radiologist's

subjective judgment, while those trained on biopsy outcomes target a more objective ground truth. Conflating these two types of labels can introduce label noise and task-definition bias, as a noticeable number of cases may have discordant BI-RADS assessments and biopsy results.

III. AVAILABLE MAMMOGRAPHY DATASETS

Several datasets have been released for mammography image analysis, covering a wide range of study sizes, imaging technologies, annotation styles, and clinical contexts. As summarized in Table 3, these datasets span multiple countries and institutions, employ imaging modalities ranging from SFM and FFDM to DBT and CESM, and vary substantially in the richness of their annotations, from approximate circular ROIs to detailed contour segmentations, as well as in the availability of clinical metadata such as patient age, breast density, BI-RADS scores, and biopsy-confirmed diagnoses. Some datasets are openly accessible, while others require institutional approval; studies relying solely on proprietary data fall outside the scope of this work. A detailed description of each dataset is provided in Appendix.

This heterogeneity in annotation granularity, task definition (screening versus diagnostic labels, BI-RADS versus biopsy-confirmed outcomes), population demographics, and imaging technology directly impact the ability to train generalizable AI models, perform fair cross-study comparisons, and draw reliable conclusions about model performance. Datasets collected in different geographic regions exhibit different breast density distributions reflecting population-level differences, while annotation formats across datasets are often mutually incompatible; bounding boxes cannot recover the spatial precision of contour annotations, and descriptive anatomical region labels cannot be converted to either. These challenges motivate the development of *MammoClean*, described in the following section, which provides a unified and reproducible open-access framework for addressing this heterogeneity systematically.

IV. HARMONIZING PUBLIC DATASETS

This section presents the *MammoClean* framework and its application to three public mammography datasets. We first categorize the main sources of heterogeneity and bias that affect mammography datasets, then describe the harmonization pipeline designed to address them, and finally present a bias analysis of the harmonized datasets. To provide a structured understanding of the different types of bias encountered in mammography data, we distinguish five categories: (1) *Technical acquisition and formatting heterogeneity*, including differences in bit depth, resolution, intensity conventions, and file organization; (2) *Annotation and metadata inconsistency*, such as varying labeling schemes, missing information, and different annotation formats; (3) *Demographic and population distribution shifts*, including differences in breast density, age, and ethnicity distributions; (4) *Label-definition mismatch*, such as discrepancies between BI-RADS assessments and biopsy-confirmed diagnoses; and

TABLE 2. BI-RADS scaling for mammography assessment and corresponding interpretations.

BI-RADS	Interpretation
0	Incomplete assessment - Additional imaging needed and further evaluation required
1	Negative – No abnormalities detected
2	Benign findings (e.g., cysts)
3	Probably benign (less than 2% malignancy risk) and short-time follow-up is recommended
4	Suspicious abnormality (2-95% malignancy risk) and biopsy may be considered
5	Highly suggestive of malignancy (more than 95% risk)
6	Known biopsy-proven malignancy that requires definitive treatment (surgery/chemotherapy)

(5) *Downstream model performance disparity*, referring to differential model performance across clinically meaningful subgroups. *MammoClean* primarily addresses the first four categories through standardization, quality control, and systematic quantification, while the fifth remains an important direction for future evaluation.

These five categories are intentionally distinct in both nature and tractability, and this distinction directly shaped the design decisions behind *MammoClean*. Categories 1 and 2 represent technical and organizational heterogeneity that can be directly resolved through standardization and quality control. Critically, the harmonization steps selected for *MammoClean* were chosen to address only issues that can be corrected without irreversible information loss, ensuring that the full richness of the original data remains accessible. Any additional annotations or dataset-specific metadata not included in the unified schema can be readily retrieved from the original sources when needed. Categories 3 and 4 cannot be eliminated by preprocessing alone, as they reflect real clinical and population-level differences; *MammoClean* instead systematically documents and quantifies them, making them visible and actionable for downstream model developers. Category 5, subgroup-level performance disparity, is a property of trained models rather than of datasets in isolation, and *MammoClean* does not itself demonstrate or guarantee equitable model performance across subgroups. Rather, by producing harmonized, well-documented datasets with quantified distributional properties, it provides the necessary foundation upon which fairness-aware model development and subgroup-level evaluation can be built. This distinction between bias identification and bias mitigation is maintained consistently throughout the remainder of this paper.

We developed *MammoClean* as a publicly accessible pipeline for processing and harmonizing mammography datasets, with the primary goal of facilitating fair and reproducible comparisons across heterogeneous sources. The three datasets selected for this case study, CBIS-DDSM, TOMPEI-CMMD, and VinDr-Mammo, represent the publicly available datasets that jointly satisfy the criteria necessary for reliable investigation of distributional shifts and dataset bias: large-scale open accessibility, sufficient study volume, and rich annotations enabling characterization of findings and their clinical context. The same pipeline can be applied to any dataset, including private datasets, making *MammoClean* a general-purpose tool rather than one tied to a specific collection.

MammoClean was developed around a multi-view structure, reflecting the standard clinical setup in which radiologists assess both CC and MLO views of each breast and make decisions at the patient level rather than the image level. This design choice mirrors clinical practice directly: the label consistency check performed across views of the same laterality is precisely the ipsilateral assessment a radiologist performs during interpretation [47]. The multi-view structure ensures consistent laterality, matched views, and unified case organization. Importantly, this design does not limit the utility of *MammoClean* to multi-view studies. Single-view analyses can equally benefit from the harmonized outputs, as diagnostic labels, breast density, and BI-RADS assessments are available at the individual image level, and the preprocessing steps addressing laterality inconsistency and intensity artifacts are relevant regardless of the number of views used downstream. The primary output of *MammoClean* includes breast density classification, BI-RADS assessment, abnormality characteristics (if available), and diagnostic labels, together with patient age where available, all stored in a unified schema. The unified output schema captures the core clinical variables common across datasets: breast density classification, BI-RADS assessment, diagnostic labels, patient age, and abnormality characteristics such as finding type, and morphology when provided by the source dataset, ensuring that no clinically relevant information present in the original data is lost in the harmonization process.

A. COMMON ISSUES AND SOURCES OF HETEROGENEITY

Several factors contribute to heterogeneity in mammography datasets, which can significantly affect both the development and performance of deep learning models. Some of these sources of variation need to be mitigated through pre-processing or harmonization, while others should be explicitly preserved to ensure that models achieve robust performance across different clinical settings.

Annotation inconsistency: The annotations and descriptive information provided across different datasets vary considerably. While some can be converted into a common format, others are not directly interoperable. For example, the VinDr-Mammo dataset provides rectangular bounding-box annotations, whereas CBIS-DDSM includes detailed enclosing contours. Although bounding boxes can be generated from contours, the reverse is not possible, as detailed contours cannot be reconstructed from rectangular boxes. In contrast, TOMPEI-CMMD offers breast-level descriptors for

TABLE 3. Summary of publicly available mammography datasets. * indicates datasets with restricted access, and + denotes the year datasets initially published with a later curated version providing more precise and detailed information.

Dataset	Year	Country	# Studies (Longitudinal)	# Images	Image Acq. Mode	Diagnosis	Finding Types	Annotations	BI-RADS Score	Breast Density	Age
MIAS [36]	1994	United Kingdom	161 (×)	332	SFM	✓	Normal, Mass, Calcification, Distortion, Asymmetry	Circle around findings with coordinates of center and approximate radius	×	✓	×
CBIS-DDSM [37]	2017 (1999+)	United States	1,391 (×)	2,844	SFM	✓	Mass, Calcification, Distortion, Asymmetry with characteristics	Contours enclosing the findings	✓	✓	×
InBreast [38]	2012	Portugal	115 (Only 8)	410	FFDM	×	Normal, Mass, Calcification, Distortion, Asymmetry	Contours enclosing findings	✓	✓	×
TOMPEI-CMMID [34]	2025 (2021+)	China	1,775 (×)	3,728	FFDM	✓	Normal, Mass, Calcification, Distortion, Asymmetry, and other features with characteristics	Descriptive anatomical region annotation	✓	✓	✓
CSAW-CC [39]*	2020	Sweden	8,723 (✓)	98,788	FFDM	✓	Normal, Invasive/Non-Invasive Cancer	Segmentation mask for some of the screen detected cases	×	×	Categorical
KAU-BCMD [40]	2021	Saudi Arabia	1,416 (×)	5,687	FFDM, US	×	N/A	N/A	✓	×	✓
RSNA [41]	2022	United States, Australia	1,970 (×)	9,594	FFDM	✓	Normal, Invasive Cancer, Non-Invasive Cancer	N/A	✓	✓	✓
VinDr-Mammo [26]	2022	Vietnam	5,000 (×)	20,000	FFDM	×	Normal, Mass, Calcification, Distortion, Asymmetry, and other features	Bounding box around findings for BI-RADS higher than 2	✓	✓	✓
MMID [42]	2024	Iraq	745 (×)	745	FFDM	✓	N/A	N/A	×	×	×
EMBED [43]*	2023	United States	115,910 (✓)	3,383,659	FFDM, DBT	✓	Normal, Mass, Calcification, Distortion, Asymmetry, Invasive Cancer, Non-Invasive Cancer, Non-Breast Cancer, High-Risk Lesion, Borderline Lesion	Region of interest	✓	×	✓
OMI-DB [44]*	2020	United Kingdom	1,311,413 (✓)	6,998,298	FFDM, DBT, MRI	✓	Normal, Invasive Cancer, Non-invasive Cancer	Region of interest	N/A	✓	✓
CDD-CESM [45]	2021	Egypt	326 (×)	2,006	CESM	✓	Normal, Mass, Calcification, Distortion, Asymmetry	Segmentation masks	✓	✓	✓
BCS-DBT [46]	2021	United States	5,610 (✓)	22,032	DBT	✓	N/A	Bounding box enclosing findings	×	×	×

abnormality locations, which cannot be converted to either bounding boxes or contours. This inconsistency poses a significant limitation for the development and evaluation of advanced models, particularly those designed for localization or segmentation tasks across multiple datasets. Another example of inconsistency is the representation of breast density. VinDr-Mammo follows the BI-RADS standard, TOMPEI-CMMD records textual descriptions of density categories, while CBIS-DDSM stores them as numerical codes. Such discrepancies can be mitigated by selecting a standard framework and mapping all dataset-specific information accordingly, thereby improving harmonization.

Bit Depth and dynamic range differences: Mammography images are often stored in high bit depth to preserve subtle contrast variations that are critical for detecting small abnormalities. However, differences in acquisition protocols and storage formats result in inconsistencies in bit depth and dynamic range across datasets, which can complicate model training and evaluation. For example, the VinDr-Mammo dataset provides images with 16-bit depth, whereas the TOMPEI-CMMD dataset stores images in 8-bit depth. Such variability can lead to loss of diagnostic information and introduce dataset-specific biases. Therefore, addressing these differences through pre-processing techniques such as normalization is an essential step in dataset preparation for AI model development.

Resolution variability: Mammography images are inherently high-resolution, but their resolution can vary depending on the imaging device, acquisition view, and breast size. Most deep learning models require a fixed input resolution, which is typically achieved by resampling the images. However, naive resampling without appropriate consideration may distort the appearance of abnormalities, thereby affecting model performance. For instance, since different mammographic views can naturally have different resolutions, resampling them all to the same resolution without zero-padding can disproportionately distort one view compared to another, leading to incorrect relative sizes of abnormalities across views. Importantly, resolution variability is not limited to differences between datasets but can also occur within a single dataset. For example, the VinDr-Mammo dataset contains images with 58 different resolutions, ranging from 2812×2012 to 3580×2812 . It is noteworthy that resampling the images to lower resolution can lead to missing information of subtle abnormalities.

Laterality flipping: Some datasets suffer from laterality inconsistencies, where the laterality information in the image headers does not match the actual side of the breast [46], or the laterality is correct but the image itself has been horizontally flipped. Ensuring consistent and accurate laterality annotation is a critical step in dataset preparation, particularly for four-view image analysis. A common pre-processing approach involves horizontally flipping one breast side to achieve consistency and facilitate ipsilateral comparisons, thereby improving feature extraction [12], [13], [17]. In certain datasets these issues have been documented,

whereas in others they require careful verification. See Figure 5, where all images are from the R-CC view but exhibit different laterality, illustrating this issue.

To illustrate the impact of flipped laterality on deep learning models, we conducted an illustrative case study comparing different multi-view fusion strategies using the ResNet18 [48] architecture, one of the most widely adopted backbones for feature extraction in mammography analysis [17], [47]. We used a subset of the TOMPEI-CMMD dataset that included all four standard mammographic views and performed nine experiments based on varying fusion strategies and laterality conditions for binary malignancy classification. We note that this experiment uses a single dataset and a single downstream task; therefore, the results should be interpreted as an illustrative demonstration rather than comprehensive evidence for general conclusions across diverse settings. The dataset was organized under three laterality conditions: (1) *Raw Data*, where left and right breast views retained their original orientations but shared consistent positioning within each laterality; (2) *Consistent Laterality*, where right breast images were horizontally flipped to achieve uniform orientation across all views; and (3) *Flipped Laterality*, where a random subset of images was horizontally flipped to simulate laterality inconsistency. The fusion strategies were defined as follows: (1) Late Fusion, where feature maps extracted by ResNet18 were combined at a later stage, following the approach of [17]; (2) Cross-View Attention (Intermediate Fusion), where cross-view attention modules were inserted after layer 3 of ResNet18 to integrate information across both ipsi-lateral and contra-lateral views, inspired by [49]; and (3) Cross-View Attention (Early-Level), where the attention module was applied after layer 2 to capture more subtle inter-view relationships, restricted to contra-lateral analysis due to computational constraints. The results of these experiments are presented in Figure 7.

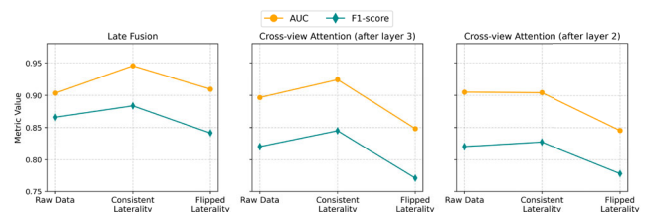


FIGURE 7. Evaluation of ResNet18-based models with different fusion strategies using raw data, data with uniform laterality, and data affected by the flipped laterality issue.

For all experiments, the dataset was partitioned at the patient level to prevent data leakage, and training was performed using the same hyperparameter settings across all conditions to ensure fair comparison. The evaluation metrics reported include Area Under the Receiver Operating Characteristic Curve (AUC), and F1-score.

In nearly all evaluations, the presence of flipped laterality led to a decline in model performance. Comparing the results obtained from raw data with those from consistently

oriented data revealed that maintaining uniform laterality improved model performance. These findings emphasize the importance of preserving a structure for image orientation, as inconsistencies, whether originating from dataset construction or data augmentation techniques, can adversely affect the learning process and overall performance of deep models. Automated detection of laterality inconsistencies can be facilitated by selecting subsets of the image from both sides and analyzing their intensity variance to identify discrepancies. Let I be a mammography image of size $H \times W$, and let n denote the window width used to extract a subset of the image from each lateral side. The laterality can then be determined by comparing the intensity variance (or standard deviation) of the left and right windows as follows:

$$\text{laterality} = \begin{cases} \text{"left"} & \text{if } \sigma(I[:, 0:n]) > \sigma(I[:, W-n:W]) \\ \text{"right"} & \text{otherwise} \end{cases} \quad (1)$$

Flipped intensity and background corruption: Standard mammography images typically have a zero-valued background, with breast structures appearing brighter relative to it. However, in some datasets this convention is not preserved, and the background may instead have the highest intensity while the breast tissue appears darker, a problem known as flipped intensity. In certain cases, flipped intensity occurs together with corrupted background sampling, further complicating image interpretation for deep models. This issue is particularly common in the VinDr-Mammo dataset and detecting and correcting such intensity abnormalities is an important step in dataset pre-processing.

To assess the effect of intensity-flipping artifacts on deep model performance, we employed the same dataset and architectures, randomly inverting image intensities for 30% and 60% of patients. The results (see Figure 8) demonstrate that increasing the proportion of intensity-flipped images leads to a noticeable degradation in model performance, highlighting the sensitivity of deep networks to intensity inconsistencies in mammography data. As with the laterality experiment, these results serve as an illustrative case study on a single dataset.

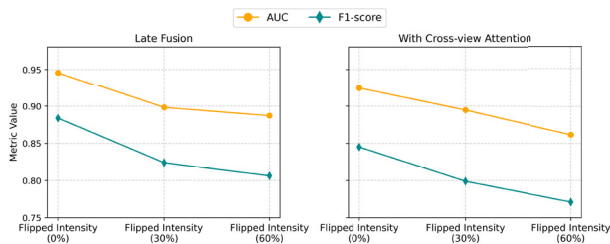


FIGURE 8. Evaluation of ResNet18-based models' performance under varying percentages of data with flipped intensity.

This issue can be detected by comparing intensity distributions in two subsets of the image, similar to the approach used for laterality detection, where the statistical features

of intensity values of predefined windows are compared. However, this method requires accurate knowledge of breast laterality to avoid misclassification. Alternative strategies, such as analyzing the image histogram, can also be applied, but these may be less robust due to variations in breast size or inter-dataset contrast differences.

Several additional factors also contribute to dataset heterogeneity, including inconsistencies in file naming and organization, missing data or metadata, differences in acquisition protocols, variability across vendors and imaging devices, and population diversity. Data curation and management practices further amplify these challenges, as some datasets are organized at the patient level while others are structured at the image level, making harmonization and unified management more complex. Addressing these issues requires dataset-specific evaluations to ensure consistency before developing a standardized framework for deep model training and validation. Missing information, such as absent views or incomplete metadata, must also be carefully managed, either by excluding incomplete studies or by adopting relevant strategies that enable model development despite missing data [11], [17], [50].

Image acquisition protocols, as well as variability across vendors and imaging devices, are additional factors that noticeably influence the visual appearance of mammography images, particularly in terms of contrast. Such shifts in contrast distribution are not only observed across datasets but also expected in real-world applications, where models must learn clinically relevant features rather than relying on dataset-specific visual patterns. Image enhancement techniques can help harmonize images from different sources by aligning their contrast distributions; however, these methods must be evaluated at large scale and remain flexible enough to adapt to diverse imaging conditions. Beyond technical variability, population diversity also plays a role in imaging characteristics; for example, breast density distributions may differ across populations, influencing the prevalence of certain conditions. Therefore, AI models should be designed and validated to perform robustly across both clinical and non-clinical sources of variability to ensure their adaptability and reliability in real-world workflows.

B. DATA STANDARDIZATION AND PREPARATION

The process of data standardization and harmonization for deep learning models can be structured into three main stages: (1) case-selection and standardizing metadata, (2) pre-processing imaging data, and (3) storing data in a unified format. Figure 9 illustrates the meta-data and imaging data process and storing in *MammoClean*.

1) INITIAL CASE SELECTION

Case selection is a process guided by the objectives of the study. Since *MammoClean* is designed to provide harmonized datasets for multi-view mammography analysis across tasks such as breast density classification, diagnostic labeling,

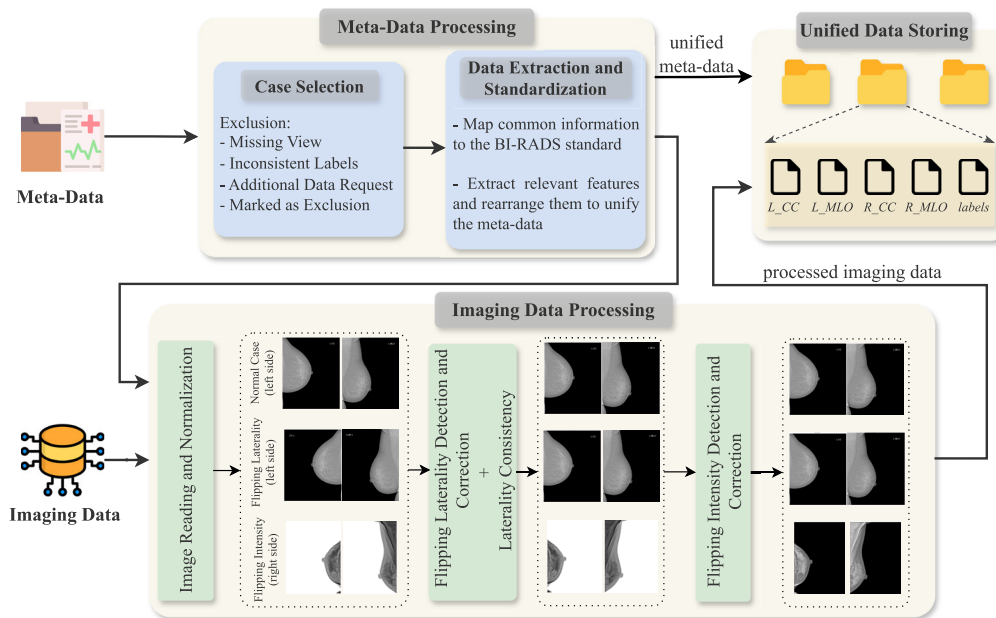


FIGURE 9. MammoClean processes heterogeneous public mammography datasets through three integrated stages: (1) metadata processing, including case selection (excluding missing views, inconsistent labels, and BI-RADS 0 assessments) and data extraction and standardization to a unified BI-RADS-aligned schema; (2) imaging data processing, comprising normalization, automated laterality flipping detection and correction, and flipped intensity detection and correction; and (3) unified data storing, where processed images and harmonized metadata are organized into a consistent multi-view structure (L-CC, L-MLO, R-CC, R-MLO) with aligned diagnostic labels.

and BI-RADS assessment, the initial data selection was performed with these goals in mind. For each dataset, the availability of complete data and the presence of potential inconsistencies between different views of the same laterality were carefully checked, and examinations with missing or inconsistent information were excluded. In CBIS-DDSM, data are organized image-based, and the inclusion and exclusion criteria applied at the examination level are illustrated in Figure 10. In TOMPEI-CMMD, imaging data are organized patient-based while metadata are provided examination-based; thus, no internal inconsistencies were observed, and only cases explicitly marked as excluded were removed from the study. In VinDr-Mammo, imaging data are organized patient-based and metadata image-based. As no critical inconsistencies were identified, all cases from this dataset were retained for analysis.

2) IMAGE DATA PROCESSING

Processing of the imaging data began with normalizing the images to ensure a consistent dynamic bit range across all cases. Subsequently, each image was checked for laterality, and to maintain consistency, images with right laterality were horizontally flipped, resulting in a uniform orientation. Following this step, an additional procedure was applied to identify images with flipped intensity and correct them accordingly. Figure 9 illustrates the overall process with representative examples. Approximately 28% of the images in the CBIS-DDSM dataset were identified with flipped

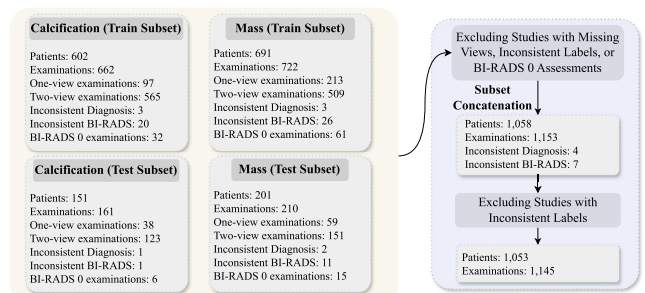


FIGURE 10. The process of case selection from the CBIS-DDSM dataset. Different subsets of images provided by the dataset were evaluated separately and then concatenated to assess possible inconsistencies. Examinations were defined on a breast basis, and those with missing views, inconsistent diagnosis labels or BI-RADS assessments, or BI-RADS 0 (requiring further imaging) were excluded. After concatenating data from different subsets, labels were double-checked to remove inconsistencies across groups due to repeated cases.

laterality issues, and about 23% of the images in the VinDr-Mammo dataset exhibited flipped intensity artifacts.

3) UNIFIED DATA STORING

To enhance reproducibility and facilitate future research, both the metadata and imaging data were unified and stored in a standardized format. Common features across the datasets were extracted and harmonized, while additional valuable features, such as the characteristics of abnormalities, even when not available in all studies, were preserved to maximize

information utility. The newly unified metadata files contain the following information:

- *ID*: Encoded patient identifier.
- *Image ID*: Encoded image identifier, if applicable.
- *Laterality*: Breast side depicted {L, R}.
- *View*: Imaging view of the breast {CC, MLO}.
- *Age*: Patient age.
- *Breast Density*: Breast density category {A, B, C, D}.
- *Diagnosis*: Biopsy-proven diagnostic label for benign and malignant cases {Normal, Benign, Malignant}.
- *BI-RADS Assessment*: BI-RADS score {1, 2, 3, 4, 5}.
- *Mass*: Presence of a mass abnormality {0, 1}.
- *Mass Shape*: Shape of the mass, if applicable.
- *Mass Margin*: Margin of the mass, if applicable.
- *Mass Density*: Density of the mass, if applicable.
- *Calcification*: Presence of a calcification abnormality {0, 1}.
- *Calcification Morphology*: Morphology of the calcification, if applicable.
- *Calcification Distribution*: Distribution of the calcification, if applicable.
- *Asymmetry*: Presence and type of asymmetry {Asymmetry, Focal Asymmetry, Global Asymmetry}.
- *Architectural Distortion*: Presence of architectural distortion {0, 1}.
- *Other Findings*: Additional abnormalities {Skin Retraction, Skin Thickening, Nipple Retraction, etc.}.
- *Split*: Dataset partition {train, test}.
- *Image File Folder (raw)*: The main folder that contains image file in the raw dataset.
- *Image File Path (processed)*: Path to the image file in the processed dataset.

It is noteworthy that, due to the presence of patient overlap between the training and test sets in the CBIS-DDSM dataset, the predefined split was discarded, and a train/test split was only retained for the VinDr-Mammo dataset. Imaging data were organized at the patient level, where each folder contained the available images along with a text file containing metadata such as age, breast density, diagnosis, and BI-RADS assessment (laterality-based, if available). The imaging files were named using the *Laterality_View* format, and this structure enables consistent reading and evaluation of the data.

C. BIAS ANALYSIS

To better analyze the datasets and identify potential biases that may influence the training and performance of deep learning models, we examined several aspects, including the distribution of data across the main classification tasks, their co-occurrence patterns, and the prevalence of abnormalities across different clinical decision-making levels. The analyzes were specifically designed around factors that directly alter the visual characteristics of mammography images in ways that go beyond well-studied sources of variability such as imaging device or acquisition protocol. Breast density was selected because it fundamentally changes the visual

appearance of the image, directly affecting the visibility of abnormalities and the difficulty of detection. BI-RADS assessment was included as it reflects the inherent ambiguity of cases from a clinical decision-making perspective, capturing the spectrum from clearly benign to highly suspicious findings in a standardized way. Together, these factors shape both what a model sees in the image and how confidently a label can be assigned, making their distributional properties across datasets a critical consideration for bias-aware dataset preparation. The reported numbers are based on the processed metadata, and some imaging files may be missing due to issues such as repository deletions or server-related problems.

Figure 11 presents the distribution of studies within the datasets for diagnostic labels, the BI-RADS scores assigned, and the distribution of these scores within each diagnostic category after harmonization using *MammoClean*. As illustrated, each dataset exhibits distinct biases toward specific conditions. In the CBIS-DDSM dataset, benign and malignant cases are relatively balanced; however, BI-RADS assessments are strongly skewed, with the majority of cases labeled as BI-RADS 4, and no representation for normal cases. In the TOMPEI-CMMD dataset, while malignant and normal cases are relatively balanced, benign cases are under-represented. BI-RADS 1 was almost exclusively assigned to normal cases, with only one benign case reported, while BI-RADS 5 represents the second most common category. In contrast, the VinDr-Mammo dataset shows a substantial imbalance, with BI-RADS 1 and 2 accounting for more than 90% of the studies. Such skewed distributions can introduce strong biases into model development, requiring careful handling; otherwise, models risk overfitting to dominant groups, and even evaluation metrics may produce misleading results.

For both TOMPEI-CMMD and CBIS-DDSM, diagnostic labels and BI-RADS scores were provided, and as expected, malignant cases were generally associated with higher BI-RADS scores compared to benign cases. Nevertheless, some discrepancies were observed: a small number of BI-RADS 5 cases had biopsy results confirming benign findings, while certain malignant cases were assigned unexpectedly low BI-RADS scores, such as 1 or 2. This highlights that although BI-RADS and biopsy results are correlated, inconsistencies exist, an issue often overlooked in studies that directly use BI-RADS categories to infer diagnostic labels.

Figure 12 shows the distribution of breast density categories across the three datasets. In CBIS-DDSM, the majority of cases fall into category B, whereas in TOMPEI-CMMD and VinDr-Mammo, category C is dominant. This shift in distribution is expected, as CBIS-DDSM was collected in the United States, while the other two datasets were collected in China and Vietnam, reflecting known differences in breast density across ethnic groups. Such distributional shifts are clinically and technically important, as higher breast densities are associated with increased difficulty in abnormality detection, potentially affecting both radiological interpretation and model performance.

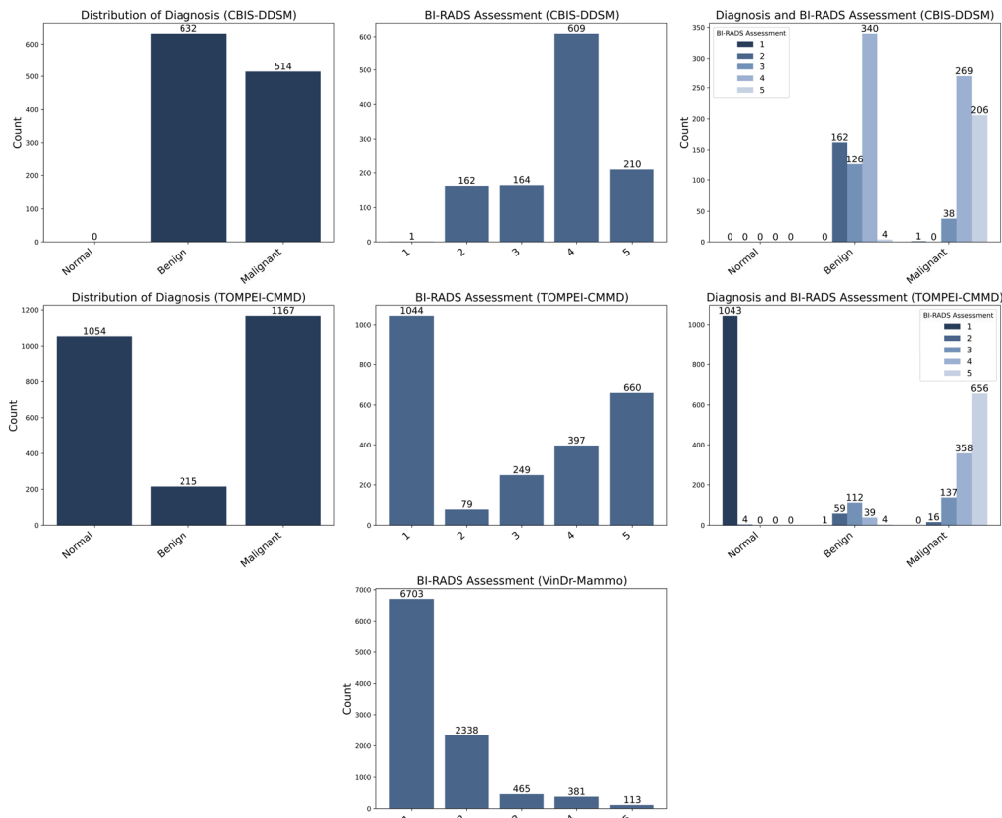


FIGURE 11. Data distribution for the diagnostic label and the BI-RADS score and their co-occurrence in different datasets after harmonization with *MammoClean*.

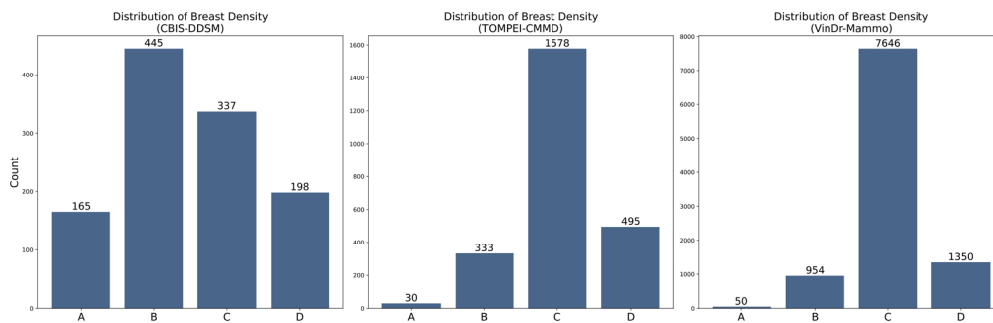


FIGURE 12. Breast density distributions across different datasets.

Table 4 provides an overview of the distribution of abnormalities across different diagnostic and BI-RADS categories in the datasets. The results demonstrate substantial variation in abnormality distributions. For instance, while mass and calcification cases in CBIS-DDSM are almost balanced, in TOMPEI-CMMD the number of mass cases is approximately double that of calcifications, and in VinDr-Mammo the ratio is closer to three. Moreover, while most calcifications in CBIS-DDSM are associated with BI-RADS categories 2 and 4 and are predominantly

benign, in TOMPEI-CMMD calcifications are primarily assigned to BI-RADS 5, with more than 86% confirmed malignant. Asymmetry and architectural distortion cases are underrepresented across all datasets, consistent with their lower prevalence in clinical practice; however, compared to CBIS-DDSM and VinDr-Mammo, these categories are even more underrepresented in TOMPEI-CMMD. It should also be noted that VinDr-Mammo does not provide annotations for BI-RADS categories 1 and 2. Overall, these discrepancies underscore the importance of adopting careful sampling

strategies to address class imbalance, a common challenge when training deep learning models with mammography data.

V. DISCUSSION

In this paper, we discussed mammography images and their analysis, beginning with the fundamentals of breast anatomy, breast density, mammographic findings, and the key factors influencing mammographic appearance, all of which can substantially affect the performance of deep models designed to automate image analysis and decision-making. We then reviewed the existing publicly available mammography datasets and observed that many of them lack the detailed findings and descriptors typically provided by radiologists during the diagnostic process. This limitation restricts the ability to evaluate model performance across subgroups defined by underlying conditions, making it difficult to identify model weaknesses and strengths and to mitigate them in order to develop fair and robust AI systems.

Another major observation is that datasets vary widely in the type and depth of information they contain, which limits their interoperability and complicates their joint use for model development. While such heterogeneity partly reflects the realities of different clinical environments, including variation in patient populations, imaging devices, and data collection protocols, it also arises from differences in data management practices, annotation strategies, and case-selection criteria, each of which can introduce its own form of bias. These differences may pose critical challenges for reproducibility, comparability, and generalizability of AI models. The distributional shifts identified across the three datasets carry direct implications for how models may behave when deployed.

To address these challenges, we developed *MammoClean*, a publicly available pipeline for pre-processing and harmonizing mammography datasets. Its primary contribution lies in the systematic integration of known harmonization needs into a unified, reproducible, and open-source framework; while individual harmonization steps are well established, no prior tool has combined them into a comprehensive, modular pipeline specifically designed for mammography. *MammoClean* was applied to three large-scale public datasets, CBIS-DDSM, TOMPEI-CMMD, and VinDr-Mammo, standardizing both imaging and metadata across heterogeneous sources and incorporating quality-control steps such as laterality verification, intensity correction, and multi-view case selection.

The distributional shifts identified across these datasets carry direct implications for model deployment. Breast density is among the most consequential factors, as it changes the visual difficulty of the task itself: a model trained predominantly on density category B cases, as in CBIS-DDSM, may encounter a systematically different decision problem when applied to populations with higher category C prevalence, as in the Asian populations represented by TOMPEI-CMMD and VinDr-Mammo. Similarly, the

BI-RADS distribution in CBIS-DDSM, where the large majority of cases fall into categories 4 and 5 with essentially no normal cases. This reflects its construction as a curated subset of abnormal findings rather than a population-level screening cohort, limiting a model's exposure to the low-suspicion cases that dominate real screening workflows. The goal of *MammoClean*'s bias analysis is not to demonstrate these failures directly, but to make the conditions for them visible and documentable so that researchers can design sampling and evaluation strategies accordingly.

Among the harmonization steps implemented, laterality correction and intensity normalization address issues that, as our illustrative experiments demonstrate, can degrade model performance even when only a fraction of cases are affected. Multi-view models rely on consistent spatial orientation for ipsilateral and contralateral comparisons, and flipped laterality in even a subset of patients disrupts the spatial relationships these models are designed to exploit. Flipped intensity values similarly alter the feature distributions that neural networks rely on for pattern recognition. These findings suggest that quality-control steps of this kind are prerequisites for reliable model training rather than optional refinements. As with all results reported here, these experiments were conducted on a single dataset and a single downstream task, and should be interpreted as illustrative demonstrations rather than comprehensive evidence for general conclusions.

Any effort to bring heterogeneous datasets into a common schema necessarily involves trade-offs, and this tension is one of the most important conceptual issues in dataset harmonization. The most concrete example concerns annotation granularity: CBIS-DDSM provides detailed contour annotations, TOMPEI-CMMD offers descriptive anatomical region labels, and VinDr-Mammo supplies bounding boxes. No single unified format can represent all three without some loss of specificity. *MammoClean* addresses this by preserving the original dataset-specific annotations alongside the harmonized output and by restricting harmonization to operations that do not introduce irreversible information loss. Intensity normalization establishes a consistent dynamic range without destroying relative contrast; laterality correction is a deterministic transformation; case exclusions are fully documented. Any dataset-specific clinical richness not captured in the unified schema can be retrieved from the original source when needed. Researchers should nonetheless be aware that the unified schema captures only the variables common across datasets, and the full analytic value of a given dataset's annotations is best exploited by working with the original source alongside the harmonized output.

MammoClean reduces technical heterogeneity and makes distributional properties visible, but cannot eliminate the underlying population-level and clinical differences that give rise to dataset bias. Regional differences in breast density reflect genuine biological variation across ethnic groups and cannot be harmonized away; case-selection practices affect both training difficulty and evaluation validity in ways that

TABLE 4. Distribution of abnormalities across diagnosis categories and BI-RADS assessments in different datasets.

Dataset	Abnormality	BI-RADS score					Diagnosis			Total
		1	2	3	4	5	Normal	Benign	Malignant	
CBIS-DDSM	Mass	1	24	121	255	136	N/A	266	271	537
	Calcification	0	143	43	357	77	N/A	371	249	620
	Asymmetry	0	4	13	7	3	N/A	20	7	27
	Architectural Distortion	0	4	0	35	16	N/A	13	42	55
TOMPEI-CMMD	Mass	1	6	162	308	533	1	122	887	1,010
	Calcification	0	73	60	117	315	4	73	488	565
	Asymmetry	1	0	25	6	0	0	18	14	32
	Architectural Distortion	0	0	9	14	13	0	7	29	36
VinDr-Mammo	Mass	N/A	N/A	279	235	95	N/A	N/A	N/A	609
	Calcification	N/A	N/A	32	118	79	N/A	N/A	N/A	229
	Asymmetry	N/A	N/A	157	72	14	N/A	N/A	N/A	243
	Architectural Distortion	N/A	N/A	12	42	11	N/A	N/A	N/A	65

preprocessing cannot resolve. *MammoClean* also operates on the accuracy of the original annotations and cannot correct labeling errors unless they are explicitly identified. Importantly, demonstrating bias mitigation at the level of model behavior across demographic subgroups remains distinct from identifying bias at the dataset level: *MammoClean* provides the necessary foundation, harmonized, well-documented datasets with quantified distributional properties, but bias mitigation in trained models requires complementary methods including stratified sampling and domain adaptation.

These findings suggest several concrete recommendations. For model training, stratified sampling to balance density and diagnostic categories is preferable to naive pooling. For evaluation, models should be assessed separately on each dataset and across clinically meaningful subgroups, by density category, abnormality type, and BI-RADS level, rather than reporting only aggregate metrics. For result reporting, disclosing dataset composition, preprocessing steps, and known selection biases alongside aggregate performance numbers would substantially improve comparability across published studies. *MammoClean* facilitates each of these practices through its transparent processing record and unified metadata format.

Although *MammoClean* represents a step toward standardized mammography data preparation, limitations remain. While harmonization can reduce unnecessary variability, it cannot replace the need for larger and more diverse datasets, nor can it fully eliminate biases inherent in clinical practice and data collection. Furthermore, while *MammoClean* identifies and quantifies biases at the dataset level, comprehensive cross-dataset model evaluation and statistical significance testing would strengthen the empirical foundations of this area of research, and these remain important directions for future work. To further advance AI for breast cancer screening and diagnosis, we suggest the following directions.

A. SUBGROUP-SPECIFIC PERFORMANCE ASSESSMENT

Most AI models are currently evaluated only on their overall outputs, without accounting for intra-category variations

within each diagnostic group. However, considering these variations is critical for developing reliable and fair models suitable for clinical use. Subgroups can be defined not only by breast density, population characteristics, or age, but also by the underlying disease type and its associated features. Evaluating model performance across such subgroups is essential for identifying potential failure modes and for guiding the development of novel approaches, such as tailored pre-processing steps or adaptive learning strategies, that can mitigate these weaknesses.

B. TOWARD CLINICALLY ALIGNED AI DECISION-MAKING

In clinical workflow, the decision-making process follows a structured approach in which radiologists not only interpret mammographic images but also integrate additional information such as patient symptoms, age, and family history. In contrast, most AI models rely almost exclusively on imaging data. This limitation is partly due to the restricted availability of comprehensive metadata in existing datasets; however, even when such information is provided, it is rarely incorporated into model development. Another critical limitation of current AI-based approaches is their tendency to produce confident predictions without offering insight into the underlying reasoning or the specific abnormalities that informed the decision. This lack of transparency, combined with overconfident outputs and the opaque nature of deep learning models, hinders their trustworthiness and adoption in clinical practice. Ideally, AI systems should emulate the reasoning process of radiologists by identifying relevant findings and clearly explaining how these contributed to the final decision. Coupled with human-in-the-loop strategies, where radiologists can correct and guide model outputs, such approaches may not only improve model performance but also move AI systems closer to reliable integration into real-world clinical workflows.

C. DIRECTIONS FOR FUTURE STANDARD DATASETS

Currently available datasets lack sufficient longitudinal data that are well organized in a time-ordered manner with step-specific decision labels and corresponding clinical

recommendations. As a result, the development of models for risk assessment has remained limited compared to diagnostic models. Longitudinal datasets could enable dynamic modeling of breast changes over time and support the detection of subtle early-stage abnormalities, in contrast to the static approaches that rely on single-study images. Moreover, future dataset curation efforts should prioritize the inclusion of detailed lesion descriptors and richer metadata, allowing AI models to incorporate multi-source information more closely aligned with clinical workflows. Ensuring broader demographic representation is equally important to improve both the generalizability and the clinical applicability of AI systems.

Recent advances in medical AI are increasingly moving beyond unimodal image-based models toward multimodal frameworks that integrate diverse sources of patient information. In mammography, this trend involves combining imaging data with complementary modalities such as ultrasound, MRI, or digital breast tomosynthesis, as well as with clinical metadata including age, family history, genetic risk factors, and prior imaging studies [51]. Multimodal integration can enhance diagnostic accuracy by capturing both morphological and contextual cues that are often considered by radiologists during routine decision-making, thereby narrowing the gap between AI systems and clinical reasoning.

A parallel development is the emergence of large-scale foundation models, pre-trained on massive heterogeneous datasets and adaptable to downstream clinical tasks [24]. These models exhibit promising transfer learning capabilities, enabling them to generalize across imaging devices, populations, and institutions. For breast cancer screening, foundation models that integrate multimodal inputs offer an avenue for more robust risk assessment, early detection of subtle changes across longitudinal exams, and alignment with clinical practice guidelines. Furthermore, the combination of foundation models with multimodal harmonized datasets, such as those enabled by *MammoClean*, may accelerate the creation of scalable and interoperable AI systems.

Nevertheless, these advances also raise critical questions about computational cost, fairness, and interpretability. While multimodal models can reduce reliance on imaging alone and provide richer diagnostic insights, they may exacerbate disparities if auxiliary metadata are incomplete or systematically biased across subgroups. Similarly, foundation models demand rigorous evaluation to ensure that their broad adaptability does not compromise specificity or introduce spurious correlations. Addressing these challenges will require collaborative benchmarking, standardized evaluation protocols, and frameworks that integrate transparency and uncertainty quantification into model outputs.

VI. CONCLUSION

In this work, we provided a comprehensive review of mammography analysis, beginning with the fundamentals of breast anatomy, breast density, and mammographic

findings, followed by an evaluation of publicly available datasets and their challenges, and introduced *MammoClean*, a reproducible and extensible pipeline for harmonizing mammography data. By standardizing imaging formats, metadata structures, and ensuring quality control through steps such as laterality verification, intensity correction, and multi-view label consistency, *MammoClean* resolves inconsistencies and unifies diverse datasets, as demonstrated on CBIS-DDSM, TOMPEI-CMMD, and VinDr-Mammo. Our comparative analysis highlighted critical insights, including regional differences in breast density distributions, discrepancies between BI-RADS assessments and biopsy-confirmed diagnoses, and imbalances in diagnostic categories that necessitate bias-aware training and evaluation strategies. While *MammoClean* reduces unnecessary variability, the need for larger, more diverse datasets and careful task design remains essential. Overall, this study offers both a comprehensive overview of mammography analysis and a methodological contribution that lays the foundation for more consistent, reproducible, and clinically relevant AI applications, advancing the long-term goal of developing equitable and reliable tools for breast cancer screening and diagnosis.

APPENDIX DETAILED DATASET DESCRIPTIONS

This appendix provides a detailed description of each publicly available mammography dataset reviewed in this work, including its collection context, imaging modality, case composition, annotation format, and available metadata. The descriptions are intended to complement the summary presented in Table 3 in the main text and to serve as a reference for researchers selecting datasets for specific analytical or modeling tasks.

MIAS: The Mammographic Image Analysis Society (MIAS) digital mammogram database consists of 322 8-bit digitized film-screen mammograms from 116 patients, all acquired in the MLO view [36]. The images were collected from a single center in the United Kingdom and annotated by expert radiologists for breast density as well as for findings in the following categories: calcifications, well-defined/circumscribed masses, spiculated masses, other masses, architectural distortion, asymmetry, and normal cases. Each abnormality was approximately localized using a circular annotation, specified by the coordinates of the center and the corresponding radius (in pixels). In addition to lesion localization, severity and diagnostic assessments were provided, classifying cases into benign or malignant groups. Breast density was assigned to three categories, fatty, fatty-glandular and dense-glandular, and no patients from the extremely dense subgroup were included in this dataset.

INBreast: The INBreast dataset [38] was collected in Portugal between 2008 and 2010 and consists of 14-bit FFDM images with resolutions ranging from 3328×4084 and 2560×3328 pixels. The dataset contains 115 cases originating from screening, diagnostic, and follow-up studies.

For 90 cases, images of both breasts were provided, including CC and MLO views (a total of four images per case). For the remaining 25 cases, only two views of a single breast were available. In addition, follow-up examinations were included for 8 cases. All abnormalities were annotated by a specialist and verified by a second expert. Annotations were provided as detailed contours delineating calcifications, masses, asymmetries, distortions, and the pectoral muscle (only for MLO view). Additional metadata included patient age, medical reports, breast density assessments, and BI-RADS categories. For 56 cases with BI-RADS scores greater than 2, biopsy results were available: 11 were confirmed as benign and the remainder malignant. The remaining cases were considered benign without biopsy confirmation. However, biopsy results are not provided in the available version at the time. However, it should be noted that the biopsy results and the patient age are not included in the currently available public version of the dataset.

DDSM and CBIS-DDSM: The Digital Database for Screening Mammography (DDSM) [52], [53] consists of 2,620 screen-film mammography studies, each containing four images, for a total of 10,480 images. The dataset includes normal, benign, and malignant cases, with associated metadata such as patient age, breast density, and BI-RADS assessment. Lesion annotations were provided as approximate regions of interest (ROIs), indicating the general location of abnormalities.

Due to limitations of the original dataset, including the relatively imprecise lesion annotations [32] and the outdated SFM format, the Curated Breast Imaging Subset of DDSM (CBIS-DDSM) [37] was later developed. In this curated dataset, a subset of DDSM cases was re-annotated by specialized radiologists to improve annotation accuracy and to review questionable cases for lesion visibility. CBIS-DDSM includes 753 cases with calcifications and 891 cases with masses, with some overlap where both findings are present in the same case. The dataset is partitioned into training and test sets across four categories based on abnormality type (mass or calcification). Despite this data arrangement, asymmetry and distortion abnormalities were also provided in the mass group. Additional metadata provided in CBIS-DDSM includes the number of abnormalities per image, mass shape, mass margin, calcification type, calcification distribution, BI-RADS assessment, pathology results (benign, malignant, and benign without callback), and subtlety ratings for abnormality visibility. However, patient age information is not included in this subset. See Figure 5 for some sample images from this dataset.

CSAW-CC: The Cohort of Screen-Aged Women Case-Control (CSAW-CC) [39] is a population-based study of women aged 40–74 years in Sweden. FFDM data were collected from three breast centers between 2008 and 2015, resulting in a dataset of 499,807 women with a total of 1,182,733 images. The complete dataset is not publicly available, but a restricted subset can be accessed, which includes 8,723 women, of whom 873 were diagnosed

with breast cancer during the observation period while the remaining served as controls. Most participants underwent more than one examination, and no examinations performed after diagnosis were included. For cancer-diagnosed cases, all prior examinations were assigned the same label, with the time between screening and diagnosis reported in three categories: less than 60 days from screening to diagnosis corresponding to screen-detected cases, 60 to 729 days corresponding to interval cancers, and more than 730 days corresponding to prior studies.

Other available information in the dataset includes cancer laterality for patients diagnosed with breast cancer, as well as cancer type, which is categorized into three groups: in situ, invasive tumors smaller than 15 mm, and invasive tumors larger than 15 mm. Patient age is provided in two categories, 40–55 years and older than 55 years, and lymph node status is reported as a binary variable indicating the presence or absence of metastasis. An important feature of this dataset is the documentation of radiologists' decisions, including whether a subject was considered healthy, required further discussion, or was recalled, along with the assigned recall label. Additional information is also available regarding the reason for recall, distinguishing between cases based on radiological findings in the images and those prompted by clinical symptoms.

CMMD and TOMPEI-CMMD: The Cancer Mammography Database (CMMD) [33] is a large-scale dataset consisting of FFDM images from 1,775 patients collected from 2012 to 2016. While some patients have all four standard views, others include only two views from a single breast. The available metadata includes biopsy results with cancer subtype information, the type of abnormality (mass, calcification, or both), and patient age. However, the dataset does not provide annotations regarding the location of abnormalities or their morphological characteristics.

The updated version, TOMPEI-CMMD [34], addresses several limitations of the original dataset and introduces additional annotations. Labeling errors involving laterality (right/left breast) and view type (MLO/CC) were corrected to improve dataset accuracy and some cases with non-visible lesions were excluded. For cases with masses or calcifications, both lesion characteristics and locations were annotated, along with the number of abnormalities present. Furthermore, additional abnormality types such as asymmetry and architectural distortion were included. Lesion locations were annotated in a descriptive, breast-based manner, categorized as lower region, medial side, middle to lateral, subareolar region, upper region, upper-medial region, and entire region. BI-RADS assessments and breast density information were also added. See Figure 6 for some sample images from this dataset.

KAU-BCMD: The King Abdulaziz University Breast Cancer Mammogram Dataset (KAU-BCMD) [40] was collected between 2019 and 2020 and includes 1,416 cases, each annotated with BI-RADS scores that were reviewed by three radiologists. For 205 of these cases, corresponding ultrasound

(US) images are also available, similarly annotated with BI-RADS assessments. While BI-RADS category 2 represents the largest group in the metadata, the currently available public version of the dataset does not include mammograms from BI-RADS 2 cases or the ultrasound images.

RSNA: This dataset was collected from sites in the United States and Australia and was released as part of a Kaggle competition [41]. It is divided into training and test subsets, with the test set containing limited metadata. The dataset is based on screening FFDM images, with most cases including all four standard views, although some have missing views. Biopsy-verified labels for cancer presence are provided, and in cases of confirmed cancer, information regarding invasive versus non-invasive subtype is also included. Since the dataset originates from screening examinations, BI-RADS scores are limited to categories 0, 1, and 2, with higher categories not represented.

VinDr-Mammo: This dataset was collected from two primary hospitals in Hanoi, Vietnam [26] and contains four-view FFDM images from 5,000 patients. See Figure 3 for some sample images from this dataset. All images were independently reviewed by two radiologists, and in cases of disagreement, a third radiologist provided the final decision. The annotated findings include masses, suspicious calcifications, asymmetry, focal asymmetry, global asymmetry, architectural distortion, skin thickening, skin retraction, nipple retraction, and suspicious lymph nodes. Each breast image was assigned a BI-RADS score and a breast density category. For non-benign findings (BI-RADS score > 2), bounding box coordinates were provided to localize the abnormality. It is important to note that biopsy-confirmed pathology results and detailed characteristic of findings such as mass and calcification are not available in this dataset.

MMD: The Mammogram Mastery Dataset (MMD) [42] was collected from four clinical units in Iraq and contains mammography images from 745 patients. For each case, either a CC or MLO view was provided, and the dataset was divided into two groups: cancer and non-cancer. In addition to the original collection, an augmented version of the dataset was released, generated using multiple image transformations such as rotation, flipping, affine transformations, noise addition, and elastic deformation. No additional clinical information or annotations were included with the dataset. It is important to note that while augmentation can improve data diversity for training purposes, the use of elastic transformation may alter key visual features that serve as indicators of malignancy, and therefore, the augmented version should be employed with caution.

EMBED: The Emory Breast Imaging Dataset (EMBED) [43] was collected between 2013 and 2020 from four institutions in the United States. It includes all women aged over 18 years who had at least one mammogram archived during this period, resulting in a total of 115,910 patients and 3,383,659 images. In addition, approximately 40,000 linked ROI annotations for lesions are provided. One of the key aims of developing this dataset was to

address the limitations of existing resources regarding race and ethnicity imbalance, and it therefore contains a more diverse population distribution, representative of the patient demographics across the contributing institutions.

The dataset comprises both screening and diagnostic studies, though the majority of the ROI annotations are associated with screening examinations. Images include FFDM, DBT, and synthesized 2D mammography, with about 42% of the examinations containing both FFDM and DBT. A comprehensive set of imaging descriptors has been applied to characterize findings, including mass (shape, margin, density), calcifications (morphology and distribution), and other abnormalities, along with detailed size and positional information. Pathology reports were also used to categorize abnormalities into seven groups reflecting severity: invasive cancer, in-situ (non-invasive) cancer, high-risk lesions, borderline lesions, non-breast cancers, and normal findings.

OMI-DB: The Optimam Mammography Image Database (OMI-DB) is a large-scale database that was collected from over 465,000 women across multiple sites in the United Kingdom and includes 1,311,413 studies with a total of 6,998,298 processed and unprocessed mammography images, spanning more than 10 years beginning in 2008 [44]. Each case contains FFDM, DBT, or both, and for a subset of high-risk cases, MRI is also available. Images are labeled as malignant, benign, interval, or normal. In addition to clinical, surgical, and pathological data available for all cases, a subset of images includes expert-provided ROI annotations. Longitudinal data and temporal linkage are also available for patients who underwent repeated studies during the collection period. The dataset includes images acquired from a range of imaging devices, including those from Hologic, Siemens, GE Healthcare, and Philips. Additional metadata includes cancer type (invasive or in-situ) and patient age. Although the dataset is accessible under restricted conditions, certain information, such as breast density, requires additional approval.

CDD-CESM: This dataset was collected in Egypt and consists of contrast-enhanced spectral mammography (CESM) images from 326 patients aged 18 to 90 years [45]. CESM was performed using standard mammography equipment following the injection of a contrast agent, with images acquired at two different energy levels. The low-energy images closely resemble conventional digital mammography, whereas the high-energy exposures capture functional activity based on contrast enhancement. For each patient, both the low-energy images and the subtracted images (where abnormalities are more clearly highlighted) are provided. Most patients have eight images available; however, for some patients, only four images from a single breast are included. In certain cases, views are missing due to low image quality or unavailability. In addition to imaging data, the dataset provides clinical information such as patient age, breast density, biopsy-verified results, BI-RADS assessments, abnormality characteristics, and radiology reports.

BCS-DBT: This dataset comprises 22,032 DBT images from 5,060 patients, collected at the Duke Health System between 2014 and 2018. It includes a total of 5,610 studies, with some patients contributing more than one study [46]. Each study contains at least one view, though the number of views per study varies. The dataset was organized into four groups based on BI-RADS assessments and pathology outcomes: (1) normal studies, defined as those with BI-RADS 1 assessments in radiology reports; (2) actionable studies, in which radiologists recommended additional imaging; (3) benign studies, selected from BI-RADS 4 assessments that underwent biopsy and were confirmed benign; and (4) cancer studies, selected from BI-RADS 4 and 5 assessments that underwent biopsy and were confirmed malignant. For the actionable, benign, and cancer groups, the initial case selection was limited to those with reported mass or architectural distortion abnormalities, while calcification cases were excluded due to their distinct visual characteristics.

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